

RC Autonomous Car: Budget and Options

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The ask

One 1:10-scale autonomous car on the RoboRacer (F1TENTH) standard: **\$1,550 now**, a complete car on the lidar and camera we already own, plus **\$400 for the lidar upgrade** when funds allow (\$1,950 together); \$3,250 for the official reference build. Every item is listed with its shop link in Section 3. The FSAE car is on a two-year cycle and cannot host autonomous work this season. The RC car is where the team develops, tests and learns in the meantime, and the software pipeline built on it is the one that goes onto the FSAE car next year.

1 What the car is and how it drives

Part	Role
Chassis	Traxxas Fiesta ST Rally VXL, ready to run: Slash 4x4 chassis, 4WD, brushless motor, steering servo. The chassis the current RoboRacer build guide and deck drawings are made for, and the car our simulator models.
Motor controller	VESC-type. Drives motor and servo from USB and reports wheel rotation back as speed.
Computer	Runs ROS 2: sensor drivers, perception, planning, control.
2D lidar	A range per angle, 5 to 40 times a second. Sees walls and cones; enough to drive with no map.
Camera	Colour: which cone is which. Needed for the perception work.
Joystick	Bluetooth gamepad for manual driving and the dead-man switch: the car drives autonomously only while a button is held.
Power	One 3S lithium pack drives the motor; the computer takes pack voltage through a fuse; a 5 V converter feeds lidar and camera.
Deck	Laser-cut plates and standoffs from the published drawings.

Flow: lidar, wheel odometry and camera in; perception locates walls or cones; a planner picks a bearing and speed; a controller turns them into steering and throttle; the VESC actuates, gated by the dead-man. Same nodes and topic names as the team's simulator, so a stack developed there drops onto the car with only the driver nodes swapped.

2 Budget

Item	Now	With S2	Reference
Chassis, batteries, charger, bag	560	560	560
Computer: Jetson Orin Nano Super, NVMe, display plug, antenna, cable	522	522	522
Motor controller: Flipsky Mini FSESC 4.20 / VESC 6 MkVI, cables, adapters	112	112	276
Power: RoboRacer powerboard, pigtail, balance extender	164	164	164
Joystick	30	30	30
Lidar: RPLidar A1M8 (owned) / RPLidar S2 / Hokuyo UST-10LX	0	399	1,200
Camera: RealSense D435i (owned) / RealSense D435i	0	0	334
Deck (cut on campus), standoffs, fastener kits	101	101	101
Spares allowance	60	60	60
Total	1,549	1,948	3,247

Prices are September 2026 shop prices (Section 3). **Now** is the official build with the sensors we own and a cheaper motor controller: a complete car that drives laps at 1 to 2 m/s. **With S2** adds the time-of-flight RPLidar, which bolts on later without any other change; it is the one upgrade that matters. **Reference** is the official RoboRacer bill of materials at today’s prices (NVIDIA raised Jetson prices by 60 to 100% in July 2026, which is why the computer costs \$399, not \$249).

3 Exact bill of materials

Every line links to the shop page. Prices are what the pages showed on 24 September 2026 (Amazon prices from the official RoboRacer sheet of May 2026, which we checked against the listings). “Now” is the car we can build today; “S2” is the lidar upgrade line. The official sheet: [RoboRacer master BOM](#).

Item	Shop	USD
Chassis and batteries		
Traxxas Ford Fiesta ST Rally VXL, 74276-4 (backordered at some dealers; Amazon B0D5M8D2R8 also lists it)	Traxxas	429.95
Zeee 3S 6000 mAh 80C hard case, EC5, 2-pack (Amazon 89.99)	Zeee	88.99
LiPo balance charger with EC5 lead	Amazon	33.99
Zeee fireproof LiPo bag	Amazon	6.99
Computer		
NVIDIA Jetson Orin Nano Super Developer Kit (backorder; Seeed 450.30; Amazon); Wi-Fi card and antennas included	SparkFun	399.00
NVMe SSD, 500 GB, M.2 2280 (sheet price; street price about 60)	Amazon	100.00
DisplayPort dummy plug (headless boot)	Amazon	6.99
7 dBi Wi-Fi antenna with MHF4 cable (range; the mount is a campus print)	Amazon	7.99
Short USB-A to micro-USB cable, Jetson to controller	Amazon	7.99
Motor controller		
Flipsky Mini FSESC 4.20 50 A (3S to 13S; flipsky.net 85, on sale 56)	Amazon	85.00
JST-PH 3-pin to header cable (controller PPM port to steering servo)	Adafruit	1.25
Header pins (three are used)	Amazon	8.99
Bullet adapters, 4 mm male to 3.5 mm female, 3-pack (Traxxas motor leads are 3.5 mm; match the controller’s bullets on arrival)	Amazon	7.99
EC5 female to XT60 male adapters, 3-pack (battery to controller)	Amazon	8.99
Power distribution		
RoboRacer power board, 1 unit (19 V to the Jetson, 12 V and 5 V rails, switch, fused; 125 each for three or more)	Ambimat	145.00
Barrel jack to pigtail, 2.5 x 5.5 mm (power board to Jetson)	Amazon	9.99

Item	Shop	USD
3S balance plug extender (battery to power board)	Amazon	8.99
Manual control		
8BitDo Ultimate 2C wireless gamepad (dead-man and manual drive)	Amazon	29.99
Sensors		
RPLidar A1M8, owned; spec	–	0
Intel RealSense D435i, owned; spec	–	0
Deck and hardware		
Platform deck, laser cut from the official DXF (3 mm acrylic sheet, campus cutter)	material	20.00
Antenna mount and A1M8 adapter plate, campus 3D printer	print	0
M3 socket-head screw kit	Amazon	22.86
M5 socket-head screw kit (two M5 x 20 are used)	Amazon	8.66
M3 hex standoffs, 14 mm	McMaster	3.68
M3 hex standoffs, 25 mm	McMaster	10.76
M3 hex standoffs, 45 mm	McMaster	27.90
M2.5 x 6 mm screws (Jetson)	McMaster	5.30
M2.5 standoffs, 10 mm (Jetson)	McMaster	2.16
Spares		
Allowance: bumper, servo saver, tyres, pinion, ordered from the Traxxas parts finder as they break	Traxxas	60.00
Now		1,549
Lidar upgrade		
RPLidar S2, 30 m time of flight (354.99 on sale today; RobotShop also stocks it)	Seeed	398.99
Now plus S2		1,948

Fitting \$1,800 with the S2. At today’s sale prices (S2 at 355, controller at 56) the total is \$1,875. Replacing the power board with a [Pololu 5 V 5 A regulator](#) (32.95) for the lidar, with the Jetson fed from the balance extender through the barrel pigtail, saves \$112 and lands at \$1,760 to \$1,835; it costs one evening of soldering and loses the fuse and switch. Cheapest of all is buying the S2 as a second, later request: the car does not wait for it.

What the A1M8 costs us. Same ROS message, same software, a printed plate instead of the Hokuyo bolt pattern, USB power instead of 12 V, so nothing about the build gets harder. What changes is the ceiling: 8,000 samples a second at 5 to 10 Hz gives about 1° resolution, ranges reliable to roughly 6 m indoors, and a scan that takes 100 to 180 ms while the car moves, so the laps are 1 to 2 m/s and the maps are coarse. Cones at 6 m give two to four returns, enough for the cone pipeline. Triangulation lidars also wash out in direct sunlight, so the A1M8 car stays indoors. The S2 lifts all of that (30 m, 32,000 samples a second, works outdoors) and is the one part that makes the car fast.

4 Options in each area

Lidar

The lidar sets the speed the car can drive: at 10 Hz it moves 25 cm between scans at 3 m/s. All publish the same ROS message, so a swap changes no code; the RPLidar sits on a small printed adapter plate and the Hokuyo bolts straight to the deck.

Lidar	USD	Rate	Range	Reality
RPLidar A1M8 (owned)	0	5 to 10 Hz	12 m	laps at 1 to 2 m/s; 1° resolution, coarse maps; runs the car until the S2 arrives
RPLidar A2M12	229	12 Hz	12 m	fine indoors; noisy on dark surfaces
RPLidar S2	399	10 Hz	30 m	time of flight; clean maps; to 3 m/s
RPLidar A3	599	15 to 20 Hz	25 m	fastest cheap unit
Hokuyo UST-10LX	1,200	40 Hz	10 m	the reference; what race winners run

Compute

Computer	USD	On the RC car	On the FSAE car
Raspberry Pi 5, 8 GB	130	lidar-only driving; no camera perception	no
Jetson Orin Nano Super, 8 GB	399 + 100 NVMe	the official part; fits the deck; enough for everything the car does	no: a 3D lidar and several cameras need more
Jetson Orin NX 16 GB on the Nano carrier	about 1,200	fits; overkill	a first driverless stack: M1 lidar, two cameras, detector
Jetson AGX Xavier 16 GB (owned, 2018)	0	no: 630 g box, does not fit the deck	a bridge for one or two seasons; last software is JetPack 5
Jetson AGX Orin 64 GB	3,500	no: 1.4 kg	the full answer, when the stack needs it

Recommended: the Orin Nano on the RC car, permanently. The FSAE computer is decided next year with a season of data; the owned Xavier carries FSAE perception development until then. If the club prefers one purchase for both cars, the Orin NX is the largest that fits the RC car and moves to the FSAE car later, at about \$800 more, leaving the RC car needing a Nano then.

Why a Jetson from day one. Jetson to Jetson is the same software: same operating system image, same ARM64 build, same CUDA and TensorRT, same ROS 2 packages. Moving the stack from the Orin Nano to an Orin NX or AGX Orin is copying it across. Developing on an x86 laptop or PC instead means a different architecture, different driver and CUDA versions, and recompiling everything, with the porting landing at the worst moment, when the FSAE car is finally ready. Building on the Nano now makes the later upgrade trivial.

A separate line for the on-car FSAE computer. If the budget process allows it, the FSAE car's computer should be its own item, requested once the team has a season of data on what the stack needs: about \$1,200 for an Orin NX 16 GB on a carrier board or about \$3,500 for an AGX Orin developer kit, at the July 2026 prices. It is not part of this ask, and buying it now would mean guessing. Two-year total, RC car with the S2 now plus the FSAE computer next year: about \$3,150 with an Orin NX, about \$5,450 with an AGX Orin.

Camera

The **RealSense D435i** we own goes on the car: colour for cone classes, depth as a bonus, and it is the camera in the official bill of materials. Nothing to buy. If a second camera is ever needed: [Logitech C920](#), \$75, rolling shutter, fine at these speeds; [OAK-D Lite](#), \$269, stereo depth and on-camera inference. [OAK-D Pro W](#) or [ZED 2i](#), \$400 to \$500, global shutter and weatherproof: FSAE-grade cameras that also fit the RC car, if the club wants the camera bought once.

Controller

The controller must accept the car's 3S pack (9 to 12.6 V). The [Flipsky Mini FSESC 4.20 50 A](#), \$85 (\$56 on sale at [flipsky.net](#)), runs 3S to 13S on VESC 4.12 hardware, the design the original F1TENTH cars raced on; USB cable included. The [VESC 6 MkVI](#), \$250, is the reference part: better cooling and support. Note that the popular Flipsky Mini FSESC 6.7 is rated 14 V minimum (4S) and does *not* suit this car. Either listed unit is a drop-in.

5 What we already own, and where it goes

- **RPLidar A1M8.** Onto the RC car: laps at 1 to 2 m/s until the S2 arrives, then a spare. USB-powered, four screws on a printed plate.
- **RoboSense M1 3D lidar.** It can be bolted to the RC car on a custom plate, and the car would drive, badly: 0.7 kg on a 3 kg car spoils the handling, its half-metre blind zone hides the nearest wall in a 2 m corridor, and its 120° forward view gives wall-following nothing to work with. It is an FSAE sensor; with the Xavier it forms a perception rig for cone-track data this semester.
- **AGX Xavier 16 GB.** Computer for that rig and a bridge for the FSAE car; not for the RC car.
- **RealSense D435i.** The RC car's camera (it is the official part); the rig borrows it when the car is idle.
- **Tamiya 1:10 chassis.** Motorless touring kit: no drawings, no match to the simulator, a semester of mechanical work to save \$300. Bench mule.
- **Alameda test track.** Outdoor sessions for the rig and, later, the FSAE car; not needed for the RC car, which runs in a corridor.

6 Why it is worth it

This semester

- The FSAE car is being built and cannot host autonomy; without a platform the team has nothing to run on until next year.
- Members stay for what they can see move.
- Real dynamics and real sensors cannot be simulated: grip, servo backlash, battery sag, a lidar that sees chair legs.
- New members learn wiring, power, safety, motor control and ROS 2 on hardware at low cost and low risk.
- The simulator and the car share one codebase, so every member's stack has somewhere to go.

Carrying over to the FSAE car

- **Same pipeline.** Cone detection with camera plus lidar, midline between cones, raceline, tracking controller: the Formula Student Driverless stack at small scale.
- **Same tools.** ROS 2 nodes and topics, sensor drivers, bag recording and replay, sim-to-real testing, safety procedures.
- **Same method.** Develop in the simulator, transfer to the car, measure the gap, close it.
- **What does not transfer:** vehicle dynamics and tuned parameters. A 3 kg car says nothing about a 250 kg car's tyres.

7 Will it work, and the practical realities

The build is the F1TENTH/RoboRacer platform, assembled by hundreds of university teams since 2016 from a public build guide; the parts list is that guide's own with the sensors we own and a cheaper controller. Driving laps is a solved problem, and our simulator already runs the same car and software. The car can run the whole pipeline: mapping, localisation, a raceline and a proper controller; the cheaper sensors cap speed, not capability. The recommended car is eligible for RoboRacer physical events as built; a lidar upgrade is what would make it fast.

- **One owner.** The car needs a named person for assembly, batteries and access, or it goes unused. This is the main risk.
- **Time.** Two to four weekends to assemble and integrate; first autonomous lap within a month; mapping, racelines and perception fill the semester.
- **Speed.** 1 to 2 m/s on the A1M8, about 3 m/s on the S2, faster only with the Hokuyo. The simulator car goes faster; members will feel the gap.
- **Access.** One car; time on it is earned through the simulator's qualification rule.
- **Batteries.** LiPo bag, supervised charging, a written procedure before the first charge.
- **Wear.** Bumpers and servo savers break; the spares line covers it.
- **Not in this budget.** A 1:5-scale car (Traxxas X-Maxx, Arrma Kraton 8S, Losi 5IVE-T: about \$3,500 even with the M1 and Alameda counted, and not runnable on campus), a second car, and the FSAE computer. All later decisions.

RoboRacer build guide: <https://f1tenth.readthedocs.io>; official master BOM: [Google Sheet](#)
Team simulator and manuals: <https://github.com/Pranman1/feb-racing>